Evaluating Behavioral Risk in AI: A Governance Framework for LLM Systems

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Executive Summary

This report presents a governance and safety audit of an LLM-based conversational system designed for entertainment recommendations with optional mood-support interaction.

While the system performs reliably in standard recommendation scenarios, this audit identifies emerging risks associated with emotionally responsive AI systems, particularly in how they influence user behavior, trust, and reliance over time.

Rather than evaluating outputs alone, this audit applies a behavioral AI governance framework to assess how repeated interaction with the system may shape cognitive and emotional patterns. Findings indicate that even appropriate responses can introduce subtle risks related to dependency, perceived empathy, and over-trust when users engage in emotionally vulnerable contexts.

This report demonstrates how structured, human-centered auditing can identify these risks prior to deployment and provides a repeatable model for evaluating AI systems beyond surface-level correctness.

System Overview

The audited system is an LLM-based chatbot designed to provide film recommendations using a movie dataset and natural language prompting.

In addition to functional recommendations, the system allows users to express emotional states such as stress, sadness, or anxiety. This creates a hybrid use case combining recommendation systems with conversational interaction that may be interpreted as emotionally supportive.

Such hybrid systems are increasingly common across consumer platforms, including streaming services, digital companions, and productivity tools. As these systems become more embedded in everyday use, governance evaluation must account not only for output quality, but for behavioral and psychological impact.

Audit Objective

This audit was designed to:

- Identify ethical, psychological, and governance risks associated with emotionally responsive AI systems
- Evaluate how system interactions may influence user behavior, trust, and decision-making
- Demonstrate a repeatable, human-centered auditing approach aligned with responsible AI principles

This reflects a shift from output-based evaluation to behavior-aware AI governance, where system impact is assessed over time rather than at the response level alone.

Framework: Behavioral AI Governance Audit Model

This audit applies a structured framework for evaluating how AI systems influence human interaction across five layers:

1. **System Interaction Patterns**

How users engage with the system over time, including frequency, reliance, and interaction depth

2. **Behavioral Signals**

Indicators such as reduced effort, automation bias, or reliance on AI-generated responses

3. **Cognitive and Emotional Impact**

Changes in attention, confidence, decision-making autonomy, and emotional response

4. **Risk Identification**

Detection of patterns such as dependency loops, over-trust, or diminished critical thinking

5. **Governance Evaluation and Recommendations**

Assessment of system alignment with human-centered AI principles and identification of design or policy improvements

This framework enables evaluation of AI systems as dynamic behavioral environments, rather than static tools.

Method

The audit used structured prompt testing to simulate real-world interaction scenarios, including:

- neutral recommendation prompts
- emotionally vulnerable or support-seeking prompts
- mixed conversational interactions

Each response was evaluated using a structured scoring framework to identify **systemic behavioral patterns**, rather than isolated outputs.

This approach prioritizes pattern detection across interactions, allowing for early identification of emerging risks.

Key Findings

- **High-risk outputs identified:** 2
- **Weak or unclear outputs:** 0
- **Transparency flags:** 0

Interpretation

The system demonstrated strong performance in functional recommendation scenarios and maintained clear system identity across most interactions.

However, responses in mood-support scenarios occasionally introduced language that could be interpreted as overly reassuring or companion-like. While not harmful in isolation, these responses signal a potential for:

- perceived emotional understanding
- increased user trust
- gradual reliance on the system for emotional support

These findings highlight the importance of evaluating not just what systems say, but how users may interpret and rely on those interactions over time.

Risk Analysis

Emotionally responsive AI systems present unique governance challenges because conversational fluency can be interpreted as empathy, authority, or understanding.

When systems provide supportive or reassuring language without clear boundaries, users may develop:

- misplaced trust
- emotional reliance
- reduced critical evaluation of responses

These risks are amplified in high-frequency interaction environments or when users engage during moments of vulnerability.

This audit demonstrates that governance evaluation must extend beyond accuracy to include behavioral and psychological impact, particularly for systems that simulate human-like interaction.

Governance Recommendations

Transparency Safeguards

Include clear system messaging that the chatbot is not a mental health professional and cannot provide therapeutic support

Emotional Boundary Design

Implement response guidelines that redirect users expressing distress toward appropriate real-world resources when necessary

Periodic Auditing

Conduct regular prompt-based audits to evaluate system behavior as models evolve

Human-Centered Evaluation

Integrate human-computer interaction review into AI deployment cycles to assess trust, clarity, and user interpretation

Business and Product Implications

Conversational AI systems increasingly influence user behavior, decision-making, and trust formation.

Governance failures in these systems can lead to:

- reputational risk

- reduced user trust and retention
- long-term product credibility issues

Conversely, organizations that embed transparency, behavioral safeguards, and human-centered design into AI systems are better positioned to build durable trust and sustained engagement.

Responsible AI governance is therefore not only a compliance requirement, but a strategic component of product reliability and long-term value.

Applications

This framework can be applied across:

- **Education:** Evaluating AI-driven learning tools and student engagement patterns
- **Workplace systems:** Assessing reliance on AI in decision-making environments
- **Consumer AI products:** Identifying dependency and engagement risks
- **Wellbeing tools:** Monitoring emotional and behavioral impact of AI interaction

Skills Demonstrated

- AI governance auditing
- Human-centered AI evaluation
- Behavioral risk analysis
- Prompt-based testing and system evaluation
- Responsible AI documentation
- Business-aligned risk assessment

Conclusion

This audit illustrates how structured, behavior-aware evaluation of AI systems can identify trust and safety risks prior to deployment.

As AI becomes more integrated into everyday environments, organizations that adopt proactive governance and human-centered design will be better positioned to build systems that users trust, understand, and use responsibly over time.